

Algebra 1
Unit 5
Lesson 2 Homework

Name Answer Key
Date _____
Period _____

1. Solve the equation and verify the solution using substitution.

$$\begin{aligned} -\frac{1}{8}y - 32 &= -\frac{3}{4}(-8 - 3y) & -\frac{1}{8}y + 7 &= 39 - \frac{3}{4}(-8 - 3y) & -\frac{1}{8}(-16) + 7 &= 39 - \frac{3}{4}(-8 - 3(-16)) \\ \frac{1}{6}y + \frac{128}{3} &= -8 - 3y & & & 2 + 7 &= 39 - 30 \\ 304 &= -19y & & & 9 &= 9 \checkmark \\ -16 &= y & & & & \end{aligned}$$

The formula for calculating the slope of a line from the points (x_1, y_1) and (x, y) is given by the following equation for questions 2-3, $m = \frac{y - y_1}{x - x_1}$.

The point-slope form of a line is as follows: $y - y_1 = m(x - x_1)$

2. Which operation (addition, subtraction, multiplication, or division) can be used to rearrange the slope formula into the point-slope form of a line?

multiplication

3. Use your answer from question 2 to rearrange the slope formula to get the point-slope form of a line.

$$\begin{aligned} (x - x_1) \cdot m &= \frac{y - y_1}{x - x_1} \cdot (x - x_1) \\ m(x - x_1) &= y - y_1 \end{aligned}$$

Use the standard form of a line is given by the formula $Ax + By = C$ for questions 4 - 5.

4. Rearrange this formula to solve for the variable, y .

$$By = C - Ax \qquad y = \frac{C - Ax}{B}$$

5. Based on your answer to question 4, what is the slope of the line you found?
How do you know? Explain your thinking in words.

The slope is $-\frac{A}{B}$. This is because the slope of any line, in the form $y = mx + b$, is m , which in this case is $-\frac{A}{B}$ since $y = \frac{-A}{B}x + \frac{C}{B}$.

For questions 6 - 7, rearrange the formulas to isolate the specified variables.

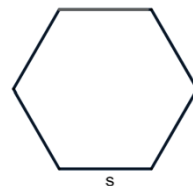
6. The formula $K = \frac{1}{2}mv^2$ is used to find the kinetic energy of an object in motion using its mass, m , and its velocity (speed), v . Solve the equation for the mass, m .

$$2K = mv^2 \qquad \frac{2K}{v^2} = m$$

7. The formula for the volume of a cylinder is given by $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height. Solve the equation for the height, h .

$$\frac{V}{\pi r^2} = \frac{\pi r^2 h}{\pi r^2} \qquad h = \frac{V}{\pi r^2}$$

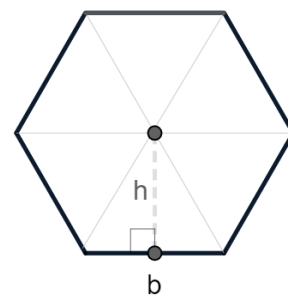
8. The formula for the area of a regular hexagon, A , is $A = \frac{3\sqrt{3}}{2} \cdot s^2$, where s is the side-length.



Lashonda rearranged the area formula to isolate the side-length, s . She knows that the area of the hexagon is 64.95. Her work is shown below. Complete the table to explain each step of Lashonda's work.

Lashonda's work:	Explanation of the steps
$A = \frac{3\sqrt{3}}{2} \cdot s^2$	Formula for the area of a regular hexagon
$2 \cdot A = \cancel{2} \cdot \frac{3\sqrt{3}}{\cancel{2}} \cdot s^2$ $2A = 3\sqrt{3} \cdot s^2$	Multiply both sides of the equation by 2
$\frac{2A}{3\sqrt{3}} = \frac{\cancel{3\sqrt{3}} \cdot s^2}{\cancel{3\sqrt{3}}}$ $\frac{2A}{3\sqrt{3}} = s^2$	Division Property of Equality
$\sqrt{\frac{2A}{3\sqrt{3}}} = \sqrt{s^2}$ $\sqrt{\frac{2A}{3\sqrt{3}}} = s$	Square Root Property of Equality
$\sqrt{\frac{2(64.95)}{3\sqrt{3}}} = s$	Substitution of Known Value
$5 \approx s$	Simplification and Rounding

Xavier is Lashonda's classmate. He said that he did not need the formula for the area of a hexagon to find the side length. He connected the vertices of the hexagon to form 6 equilateral triangles, shown in the figure to the right. He used the formula for the area of a triangle, $A_{\text{triangle}} = \frac{1}{2}bh$, to create the following formula for the area of a regular hexagon: $A_{\text{hexagon}} = 6\left(\frac{1}{2}bh\right)$.



Xavier got stuck because he did not know the height, h , of the triangles!

9. Rearrange Xavier's formula, $A_{\text{hexagon}} = 6\left(\frac{1}{2}bh\right)$, to solve for the height, h .

$$\frac{A_{\text{hexagon}}}{6} = \frac{6\left(\frac{1}{2}bh\right)}{6}$$

$$\left(\frac{2}{b}\right) \cdot \frac{A_{\text{hexagon}}}{6} = \frac{1}{2}bh \cdot \left(\frac{2}{b}\right)$$

$$\frac{A_{\text{hexagon}}}{3b} = h$$

10. Substitute the area, 64.95, and the side-length Lashonda found, 5, to find the height of Xavier's triangles.

$$\frac{(64.95)}{3(5)} = h$$

$$\frac{64.95}{15} = h$$

$$4.33 = h$$

The heights are equal to 4.33 each.